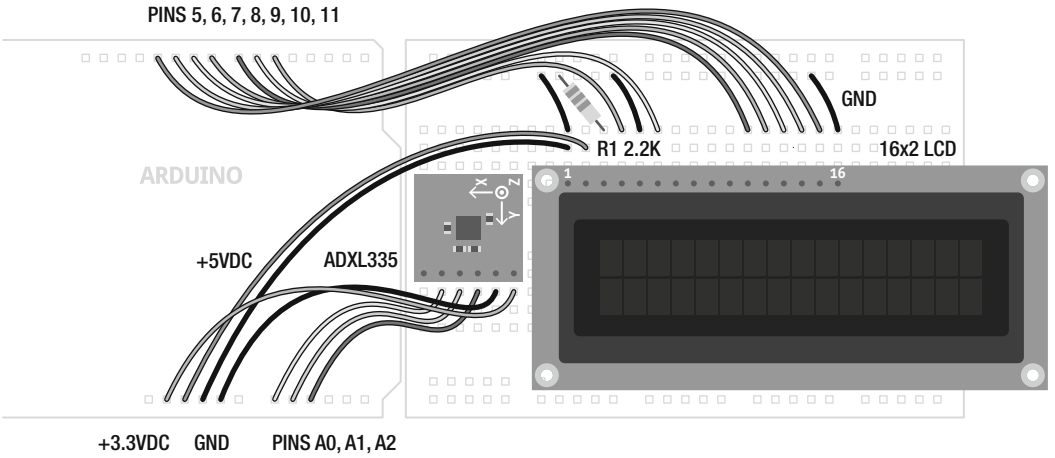


Figure 8-2. *Decision Machine illustration*

Uploading the Source Code

While you might think this sketch is a bit intimidating and lengthy, in reality it's not too bad because most of it is white space. Half of the source code is tied up in listing the 20 possible answers that could be displayed each time the device is shaken or rotated. The rest builds on prior projects to come before it by using many of the functions and structures used in past sketches. We've stuck with some answers that might seem familiar as a starting point, but you should definitely make up your own answers. To get things to line up in the center of the screen, each answer is written as two lines of 16 characters using spaces as appropriate to center each line on the screen.

To compartmentalize our code a bit, we have created three new functions. The first function, named `getReading()`, reads the three analog inputs for each of the axes on the accelerometer. The second, `oldReading()`, keeps track of the last sensor readings so that we can determine if a threshold has been crossed. Finally, `getAnswer()` generates a random answer from the 20 possible, gets the message text to be displayed, and then sends that message out to the LCD display. To create a sense of drama, this function will also fade the display up to full brightness and then back again to darkness to simulate the response bubbling up from the aether within. Before we get into how this all works, we need to wire up the circuit, upload the source code in Listing 8-1, and see what happens.

Listing 8-1. *Decision Machine Source Code*

```
#include <LiquidCrystal.h>

char* allAnswers[] = {
"  As I see it,  ",
"      yes       ",

"     It is      ",
"    certain     ",
```